Gate 02 — Data Layer

Status: Complete **Preconditions:** Gate 01 complete **Unlocks:** Gate 03 (Auth & Access Control)

Purpose

Create all Django applications and core models with migrations. This gate produces the database schema that every subsequent gate builds on. No business logic, no endpoints, no admin configuration — just models and migrations.

All BA-approved model changes from ba-assessment-v1.md are included here. Undescribed taxa are first-class from day one.

Django Applications

App	Models in This Gate
`species`	Species, Taxon, CommonName, ConservationAssessment, SpeciesLocality, Watershed, ProtectedArea
`populations`	Institution, ExSituPopulation, HoldingRecord
`fieldwork`	FieldProgram
`accounts`	User, AuditLog
`integration`	SyncJob

Apps not created in this gate (post-MVP): `coordination` (BreedingRecommendation, PrioritizationScore), `fieldwork` extensions (Survey, OccurrenceRecord).

Model Specifications

`species.Species`

Field	Type	Notes
`scientific_name`	CharField(200)	Full designation including informal suffix for undescribed taxa (e.g., "Bedotia sp. 'manombo'")
`taxonomic_status`	CharField, enum	`described` / `undescribed_morphospecies` / `species_complex` / `uncertain` — **not nullable; default `described`**
`provisional_name`	CharField(100), nullable	Informal epithet only (e.g., "'manombo'"). Null for described species.
`authority`	CharField(200), **nullable**	Null for undescribed taxa
`year_described`	IntegerField, **nullable**	Null for undescribed taxa

`family`	CharField(100)	e.g., "Bedotiidae", "Cichlidae"
`genus`	CharField(100)	
`taxon`	FK → Taxon, nullable	Link to hierarchical taxonomy
`endemic_status`	CharField, enum	`endemic` / `native` / `introduced`
`iucn_status`	CharField, enum	`EX` / `EW` / `CR` / `EN` / `VU` / `NT` / `LC` / `DD` / `NE` — nullable
`population_trend`	CharField, enum	`increasing` / `stable` / `decreasing` / `unknown` — nullable
`cares_status`	CharField, enum	`CCR` / `CEN` / `CVU` / `CLC` — nullable (null = not listed)
`shoal_priority`	BooleanField	Default False
`description`	TextField, blank	Public narrative
`ecology_notes`	TextField, blank	
`distribution_narrative`	TextField, blank	
`morphology`	TextField, blank	
`max_length_cm`	DecimalField(5,1), nullable	
`habitat_type`	CharField(100), blank	
`iucn_taxon_id`	IntegerField, nullable, unique	IUCN API identifier. Null for undescribed taxa without IUCN records.
`fishbase_id`	IntegerField, nullable	
`gbif_taxon_key`	IntegerField, nullable	
`created_at`	DateTimeField, auto	
`updated_at`	DateTimeField, auto	

`species.Taxon`

Field	Type	Notes
`rank`	CharField, enum	`family` / `genus` / `species` / `subspecies`
`name`	CharField(200)	
`parent`	FK → self, nullable	Hierarchical; use `django-mptt` or nested set
`common_family_name`	CharField(200), blank	e.g., "Malagasy Rainbowfishes"

`species.CommonName`

Field	Type	Notes
`species`	FK → Species	

`name`	CharField(200)	
`language`	CharField(10)	ISO 639-1: `en`, `fr`, `mg`
`is_preferred`	BooleanField	Default False; one preferred name per language

`species.ConservationAssessment`

Field	Type	Notes
`species`	FK → Species	
`category`	CharField, enum	`EX` / `EW` / `CR` / `EN` / `VU` / `NT` / `LC` / `DD` / `NE`
`criteria`	CharField(100), blank	IUCN criteria string (e.g., "A2acd")
`assessor`	CharField(200), blank	
`assessment_date`	DateField, nullable	
`source`	CharField, enum	`iucn_official` / `recommended_revision`
`notes`	TextField, blank	
`review_status`	CharField, enum	`accepted` / `pending_review` / `under_revision` / `superseded` — **default `accepted`**
`review_notes`	TextField, nullable	Free-text explanation of why flagged
`flagged_by`	FK → User, nullable	Coordinator who flagged
`flagged_date`	DateTimeField, nullable	
`created_at`	DateTimeField, auto	

`populations.Institution`

Field	Type	Notes
`name`	CharField(300)	
`institution_type`	CharField, enum	`zoo` / `aquarium` / `research_org` / `hobbyist_program` / `ngo` / `government`
`country`	CharField(100)	
`city`	CharField(100), blank	
`zims_member`	BooleanField	Default False
`species360_id`	CharField(50), blank	

`eaza_member`	BooleanField	Default False
`aza_member`	BooleanField	Default False
`website`	URLField, blank	
`contact_email`	EmailField, blank	Tier 3+ only

`populations.ExSituPopulation`

Field	Type	Notes
`species`	FK → Species	
`institution`	FK → Institution	
`count_total`	IntegerField, nullable	Latest known total
`count_male`	IntegerField, nullable	
`count_female`	IntegerField, nullable	
`count_unsexed`	IntegerField, nullable	
`date_established`	DateField, nullable	
`founding_source`	CharField(300), blank	
`breeding_status`	CharField, enum	`breeding` / `non-breeding` / `unknown`
`studbook_managed`	BooleanField	Default False
`last_census_date`	DateField, nullable	
`notes`	TextField, blank	

`unique_together` : `(species, institution)`

`populations.HoldingRecord`

Field	Type	Notes
`population`	FK → ExSituPopulation	
`date`	DateField	
`count_total`	IntegerField	
`count_male`	IntegerField, nullable	
`count_female`	IntegerField, nullable	
`count_unsexed`	IntegerField, nullable	
`notes`	TextField, blank	
`reporter`	FK → User, nullable	

`fieldwork.FieldProgram`

Field	Type	Notes
`name`	CharField(300)	e.g., "Fish Net Madagascar", "Durrell Nosivolo"
`description`	TextField	Public-facing summary
`lead_institution`	FK → Institution, nullable	
`region`	CharField(200), blank	e.g., "Nosivolo River catchment"
`start_date`	DateField, nullable	
`status`	CharField, enum	`active` / `completed` / `planned`
`focal_species`	ManyToMany → Species	
`partner_institutions`	ManyToMany → Institution	
`funding_sources`	TextField, blank	
`website`	URLField, blank	

`accounts.User`

Custom user model extending `AbstractBaseUser` + `PermissionsMixin`.

Field	Type	Notes
`email`	EmailField, unique	Primary identifier (no username)
`name`	CharField(300)	
`access_tier`	IntegerField	1–5; default 2 (Registered Researcher) for self-registered accounts
`institution`	FK → Institution, nullable	Primary affiliation
`expertise_areas`	TextField, blank	
`orcid_id`	CharField(50), blank	
`is_active`	BooleanField	Default False until email verified
`is_staff`	BooleanField	Default False; True for Django Admin access
`date_joined`	DateTimeField, auto	

Tier 1 = unauthenticated access (no User record required). `access_tier=1` on an authenticated User is valid but unusual.

`accounts.AuditLog`

Field	Type	Notes

`user`	FK → User, nullable	Null for anonymous actions
`action`	CharField, enum	`create` / `update` / `delete`
`model_name`	CharField(100)	e.g., "Species"
`object_id`	CharField(50)	PK of affected record
`timestamp`	DateTimeField, auto	
`changes`	JSONField	Before/after diff
`ip_address`	GenericIPAddressField, nullable	

Append-only. No update/delete signals on this model. Use a `pre_delete` signal to block deletion attempts.

`integration.SyncJob`

Field	Type	Notes
`job_type`	CharField, enum	`iucn_sync` (only value for MVP)
`status`	CharField, enum	`pending` / `running` / `completed` / `failed`
`started_at`	DateTimeField, nullable	
`completed_at`	DateTimeField, nullable	
`records_processed`	IntegerField, default 0	
`records_updated`	IntegerField, default 0	
`records_skipped`	IntegerField, default 0	
`error_log`	TextField, blank	Accumulated errors

`species.SpeciesLocality`

Each record represents a single known locality for a species — sourced from type descriptions, museum collection records, published literature, field observations, or eDNA surveys. This is the primary point layer for the conservation map.

Field	Type	Notes
`id`	AutoField	PK
`species`	FK → Species	`on_delete=CASCADE`; indexed
`locality_name`	CharField(300)	e.g., "Amboaboa River at Antsirabe confluence"
`location`	PointField(srid=4326)	Exact coordinates (WGS 84)
`location_generalized`	PointField(srid=4326), nullable	Pre-computed generalized coordinates for public display. Null if not sensitive.
`water_body`	CharField(200), blank	Name of river, lake, etc.

`water_body_type`	CharField, enum	`river` / `lake` / `stream` / `cave_system` / `wetland` / `estuary`
`drainage_basin`	FK → Watershed, nullable	Enables efficient "all species in this watershed" queries
`drainage_basin_name`	CharField(200), blank	Denormalized basin name. Auto-populated from FK on save.
`locality_type`	CharField, enum	`type_locality` / `collection_record` / `literature_record` / `observation`
`presence_status`	CharField, enum	`present` / `historically_present_extirpated` / `presence_unknown` / `reintroduced`
`source_citation`	TextField	Required. Provenance is non-negotiable for biodiversity data.
`year_collected`	IntegerField, nullable	Year of the record, not the publication year.
`collector`	CharField(200), blank	Collector or observer name(s).
`coordinate_precision`	CharField, enum	`exact` / `approximate` / `locality_centroid` / `water_body_centroid`
`is_sensitive`	BooleanField, default False	If True, `location_generalized` is served to Tier 1-2 users instead of `location`.
`notes`	TextField, blank	
`created_at`	DateTimeField, auto_now_add	
`updated_at`	DateTimeField, auto_now	

Constraints:

- `unique_together = (species, location, locality_type)`
- `save()` override: if `is_sensitive` and `location` is not null, compute `location_generalized` as `Point(round(lng, 1), round(lat, 1))`. If not sensitive, set `location_generalized = None`.
- `save()` override: if `drainage_basin` FK is set and `drainage_basin_name` is empty, populate from `drainage_basin.name`.

`species.Watershed`

Reference layer storing HydroSHEDS HydroBASINS polygons for Madagascar. Used as FK target for SpeciesLocality.drainage_basin and as the source for the watershed map overlay.

Field	Type	Notes
`id`	AutoField	PK
`hybas_id`	BigIntegerField, unique	HydroBASINS feature ID

`name`	CharField(200)	Basin name; fallback "Unnamed basin [hybas_id]" for unnamed features
`pfafstetter_level`	IntegerField	Pfafstetter coding level (MVP loads level 6 only)
`pfafstetter_code`	BigIntegerField	Pfafstetter basin code
`parent_basin`	FK → self, nullable	For hierarchical basin navigation
`area_sq_km`	DecimalField(12,2), nullable	Basin area from HydroBASINS attributes
`geometry`	MultiPolygonField(srid=4326)	Basin boundary polygon
`created_at`	DateTimeField, auto_now_add	

`species.ProtectedArea`

Reference layer storing WDPA protected area polygons for Madagascar. Used for the protected areas map overlay.

Field	Type	Notes
`id`	AutoField	PK
`wdpa_id`	IntegerField, unique	WDPA feature ID
`name`	CharField(300)	Protected area name
`designation`	CharField(200)	e.g., "National Park", "Special Reserve"
`iucn_category`	CharField(20), blank	IUCN PA category (Ia, Ib, II, III, IV, V, VI)
`status`	CharField(100)	e.g., "Designated", "Proposed"
`status_year`	IntegerField, nullable	Year of designation
`area_km2`	DecimalField(12,2), nullable	Reported area
`geometry`	MultiPolygonField(srid=4326)	PA boundary
`created_at`	DateTimeField, auto_now_add	

Technical Tasks

- Run `django-admin startapp` for `species`, `populations`, `fieldwork`, `accounts`, `integration`
- Add all apps to `INSTALLED_APPS` in settings
- Create custom User model before first migration (cannot be changed after)
- Configure `AUTH_USER_MODEL = 'accounts.User'`
- Enable PostGIS: `django.contrib.gis` in `INSTALLED_APPS`; `DATABASES['default']['ENGINE'] = 'django.contrib.gis.db.backends.postgis'`
- Install `django-mptt` for Taxon hierarchy; add `MPTTModel` to Taxon

- Run initial migrations; confirm PostGIS extension enabled (`SELECT PostGIS_Version();`)
- Write model-level unit tests: field nullability, enum constraints, `unique_together` on ExSituPopulation

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Acceptance Criteria

Given a new Species record with `taxonomic_status = 'undescribed_morphospecies'` **When** `authority` and `year_described` are set to null **Then** the record saves without error and no null constraint is violated

Given a new Species record with `taxonomic_status = 'described'` **When** the record is saved with `provisional_name` set **Then** the record saves (provisional_name is nullable and optional for described species)

Given a ConservationAssessment record with `review_status = 'pending_review'` **When** `flagged_by` and `flagged_date` are null **Then** the record saves (both fields are nullable — flagging via Django Admin may not always record user context)

Given an attempt to set `access_tier` outside the range 1–5 on a User record **When** the record is saved **Then** a validation error is raised (enforce with `validators=[MinValueValidator(1), MaxValueValidator(5)]`)

Given two ExSituPopulation records with the same `species` and `institution` **When** the second is saved **Then** a database-level unique constraint violation is raised

Given a SpeciesLocality record with valid coordinates (-18.91, 47.52) **When** the record is saved **Then** the PointField stores the coordinates correctly; querying with `ST_Contains` against a polygon that includes those coordinates returns the record

Given two SpeciesLocality records with the same `species` , `location` , and `locality_type` **When** the second is saved **Then** a database-level unique constraint violation is raised

Given a SpeciesLocality record with `is_sensitive = True` and `location = Point(47.5234, -18.9156)` **When** the record is saved **Then** `location_generalized` is auto-computed as `Point(47.5, -18.9)` (rounded to 0.1 degree)

Given a SpeciesLocality record with `is_sensitive = False` **When** the record is saved **Then** `location_generalized` is null

Given a Watershed record with a valid MultiPolygon geometry **When** the record is saved with `hybas_id = 1060000010` **Then** the record saves without error; a second record with the same `hybas_id` raises a unique constraint violation

Given a ProtectedArea record with `wdpa_id = 303847` **When** the record is saved **Then** the record saves without error; a second record with the same `wdpa_id` raises a unique constraint violation

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Out of Scope

- Admin configuration (Gate 04)
- DRF endpoints or serializers (Gate 05)
- Auth enforcement / permission classes (Gate 03)
- Occurrence records, Survey, BreedingRecommendation, PrioritizationScore (post-MVP)

Gate Exit Criteria

Before marking Gate 02 complete:

1. All migrations apply cleanly on a fresh PostGIS database (`migrate --run-syncdb` passes)
2. All model-level unit tests pass
3. Custom User model is set before any auth migrations run
4. SpeciesLocality, Watershed, and ProtectedArea migrations apply cleanly; spatial field tests pass (PointField and MultiPolygonField accept valid geometries)
5. Unique constraints enforced on `hybas_id` , `wdpa_id` , and `(species, location, locality_type)`
6. `save()` override on SpeciesLocality correctly computes `location_generalized` for sensitive records and populates `drainage_basin_name` from FK
7. Invoke **@code-quality-reviewer** on all model files

Reconciliation

Gate 02 implementation was reconciled on 2026-04-16. Full reconciliation report: gate-02-reconciliation.md

Key deviations from spec (all justified, none blocking):

- SpeciesLocality unique constraint uses deterministic `location_key` CharField instead of raw geometry equality
- CheckConstraint uses `condition=` (Django 5.1+ forward-compatible) instead of deprecated `check=`
- Circular accounts/populations migration dependency resolved via split 0001/0002 migration
- `drainage_basin_name` always syncs from FK on save (not conditional on empty)
- SyncJob.error_log implemented as JSONField instead of TextField
- FieldProgram.end_date added (not in original spec)